

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices $A_1, A_2, \dots, A_r$.

The tensor T is shown as a purple cube, labeled T , with dimensions $\in \mathbb{R}^{n \times m \times p}$.

The decomposition is represented by the equation:

$$T = \sum_{i=1}^r A_i \cdot G_i$$

where the components are:

- A_1 : A blue vertical line segment.
- A_2 : A blue vertical line segment.
- $\dots$: Ellipsis indicating intermediate terms.
- A_r : A blue vertical line segment.
- G_1 : A green diagonal line segment.
- G_2 : A green diagonal line segment.
- $\dots$: Ellipsis indicating intermediate terms.
- G_r : A green diagonal line segment.

The core tensor G is represented by the sum of the core tensors G_i (green diagonal line segments) and the factor matrices A_i (blue vertical line segments).